

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
Mid-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
8th October 2024

Time: 2 hours

Total Marks: 20

Answer all questions.

Q 1. Find the interval in which the positive roots of the equation $x - e^{-x} = 0$ lie. Find the smallest positive roots of the equation using Regula-Falsi method. 5

Q 2. Find the solution of the system of equations

$$4x_1 + x_2 + x_3 = 4$$

$$x_1 + 4x_2 - 2x_3 = 4$$

$$3x_1 + 2x_2 - 4x_3 = 6$$

using the Gauss elimination method. 5

Q 3. Using Lagrange's formula, find the interpolating polynomial $P(x)$ of degree 2 or less that takes the following values. 10

$$x_0 = 1, P(x_0) = 1$$

$$x_1 = 3, P(x_1) = 27$$

$$x_2 = 4, P(x_2) = 64$$